

Robot Challenge & Field

Spring 2025

Mechatronics Laboratory (Robot: Advanced)

Scoring Details (for each stage)

Total Scores = 140 points

= 100 points + 40 points

Points from field

Based on the robot performance in the field

Points from Instructors

Average of all instructor's points based on:

1. Originality
2. Presentation
3. Robot Completeness
4. Team Effort

General Rules

1. Robots can be **autonomous**, **teleoperated**, or **hybrid**!
2. Robots should be **untethered**. (no cable connection for power, etc.)
3. A team member can help the robot (called **intervention**) during the attempt if it gets stuck or lost.
 - **Each intervention will entail penalty** (reduce points).
4. Use of **ready-to-use robot kits is NOT allowed**.
 - Teams are expected to build their robots by themselves as much as possible.
 - Only essential components (e.g., motor, wheels, electrical components, etc.) can be purchased and used.

For Teleoperated Robots:

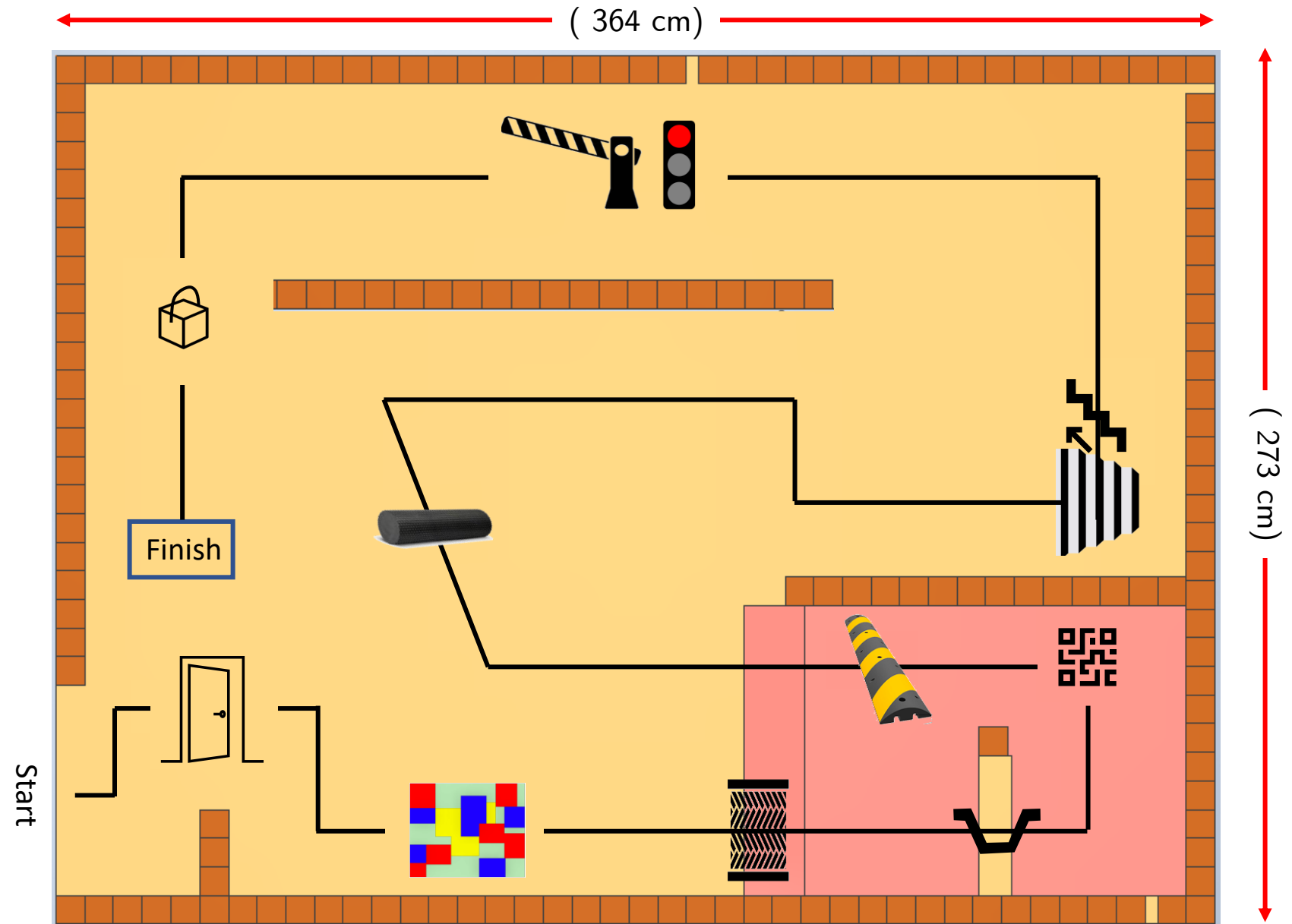
- Operator cannot see the robot directly in the field. They must operate remotely from a fixed location (**operator station**) using video from the on-field camera.
- **Optionally**, you can add additional cameras or other sensors on the robot for further support during the teleoperation. In that case, **you have to setup a wireless master-slave ROS connection**.
- We will **provide a dedicated wifi router** (ID: ■■■■, pwd: ■■■■■■) as *kuas-wlan* does not allow such a connection.

Robot Challenge Field

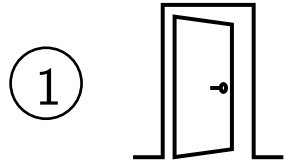
Black line width: 20 mm

Elevation

	0 cm		3 cm
	0-3 cm		3-10 cm



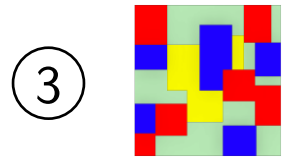
Robot Challenge - Task Details (Stage-I)



Door: open it in any way possible



Slope: Climb up the slope



Uneven Terrain: traverse uneven terrain or random steps



Ditch Crossing: Cross the ditch



QR Code: Read and change the color of an on-board LED

⑥



Bump: cross the bump

⑦



Roller: cross the pipe

⑧



Stairs: can climb up/down from any direction

⑨



Traffic light & Barrier: detect light status and open the barrier using wireless communication

⑩



Obstacle: Transport the object to goal point by any means

Task Points

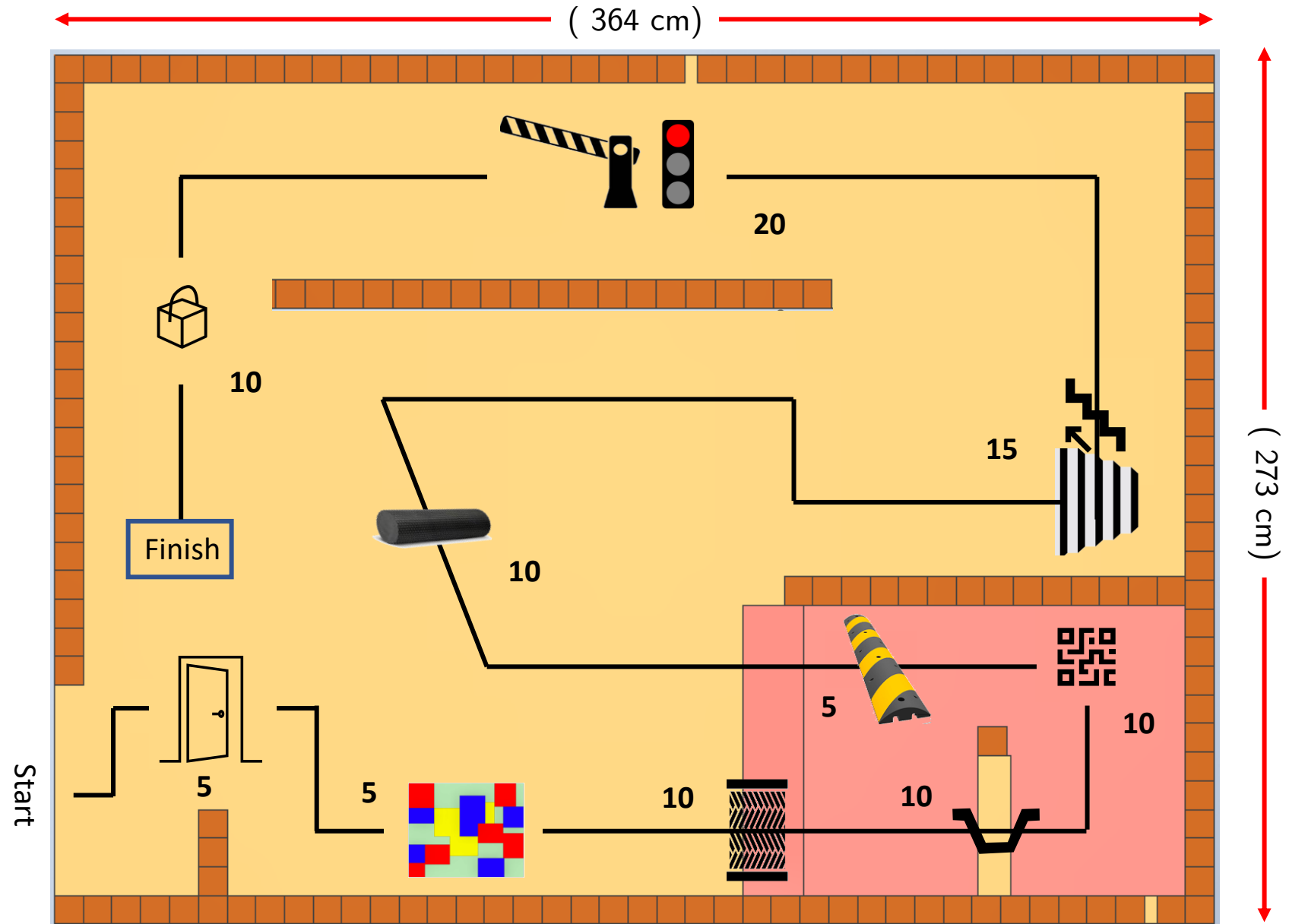
Stage-I

- Digits represent the points for each task
- **Stairs:** 8 points for successful climb, 7 for coming down

Number of Interventions
allowed:

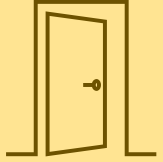
Autonomous: 5 times
Penalty: -5 points per
intervention

Teleoperated: 3 time
Penalty: -10 points per
intervention



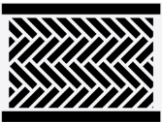
Robot Challenge - Task Details (Stage-II)

①



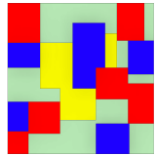
Door: open the door using knob

②



Slope: Climb up the slope

③



Uneven Terrain: traverse uneven terrain or random steps

④



Ditch Crossing: Cross the ditch

⑤



QR Code: Rotate the cardboard, Read and change the color of an on-board LED

⑥



Bump: cross the bump

⑦



Roller: cross the pipe

⑧



Stairs: can climb up/down from any direction

⑨



Traffic light & Barrier: detect light status and open the barrier using wireless communication

⑩



Obstacle: Lift the object and transport to the goal point

Task Points

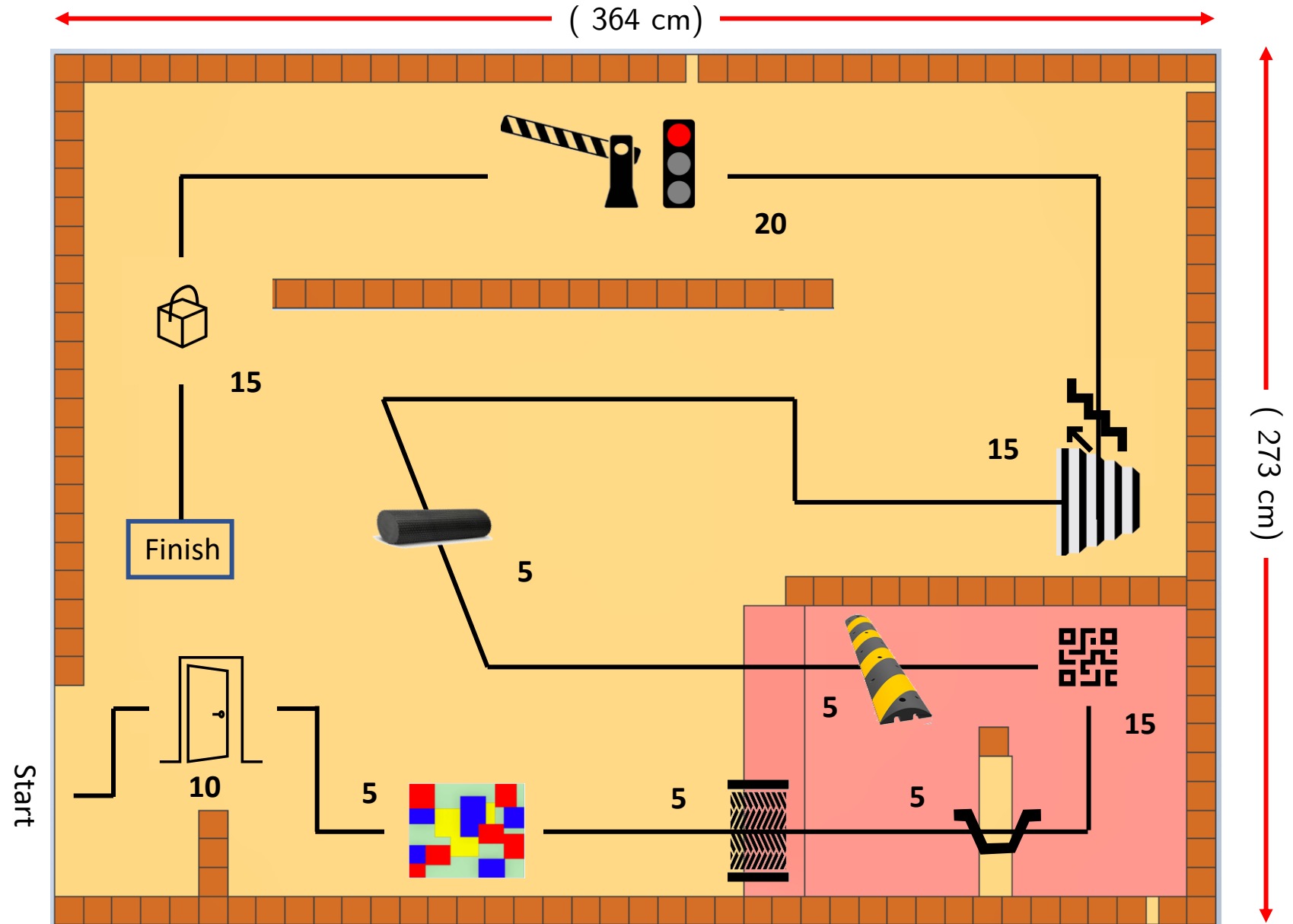
Stage-II

- Digits represent the points for each task
- **Stairs:** 8 points for successful climb, 7 for coming down

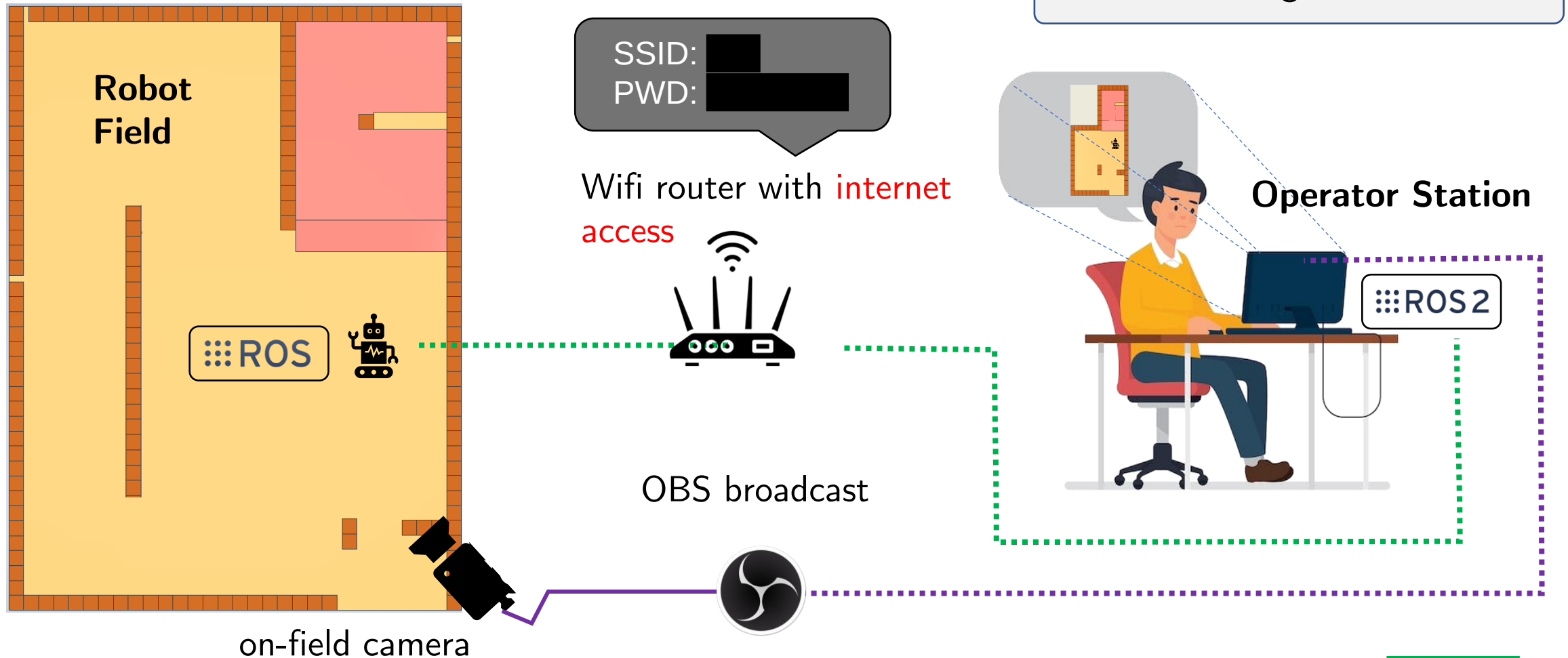
Number of Interventions
allowed:

Autonomous: 5 times
Penalty: -5 points per
intervention

Teleoperated: 3 time
Penalty: -10 points per
intervention

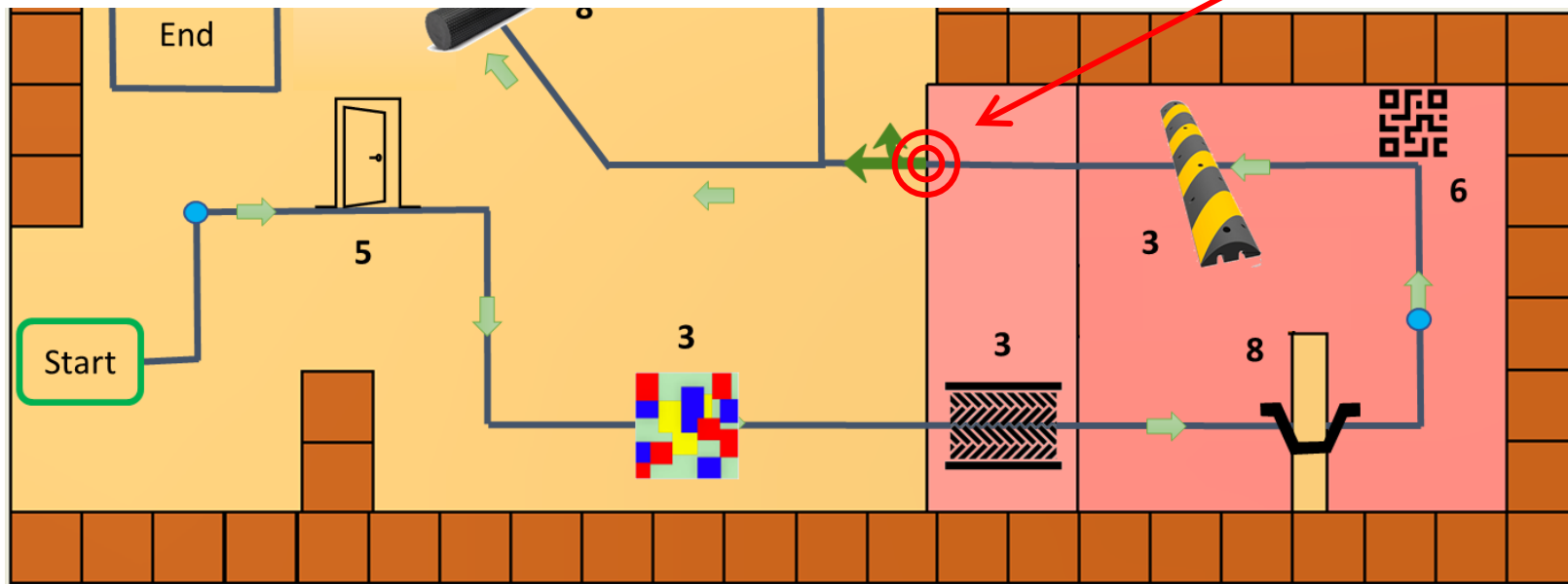


Teleoperated Robots Operation



For Autonomous Robots:

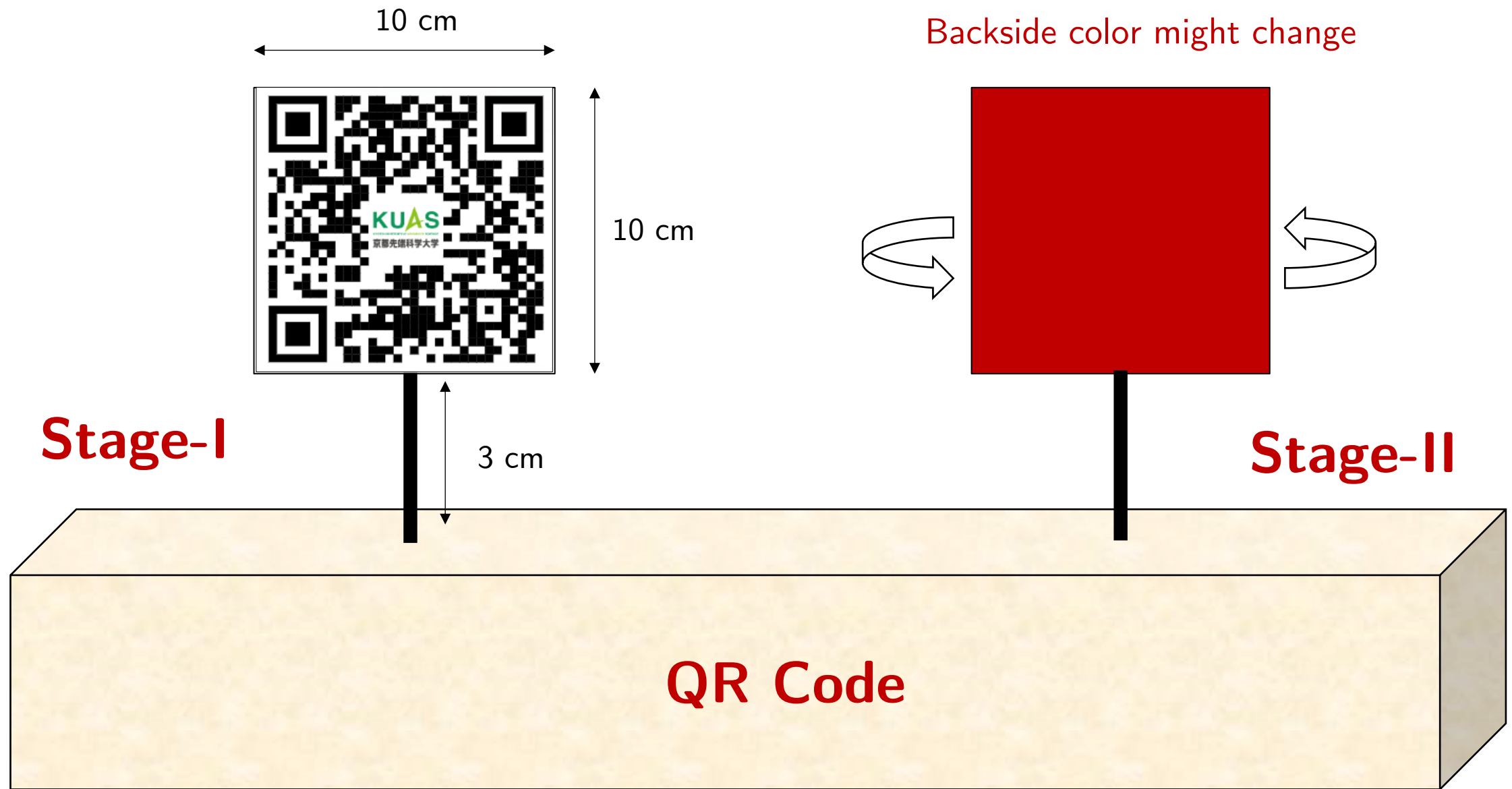
- **Autonomous Robots** will get 15 additional points compared to the teleoperated counterparts if they can reach successfully to the **marked point**.



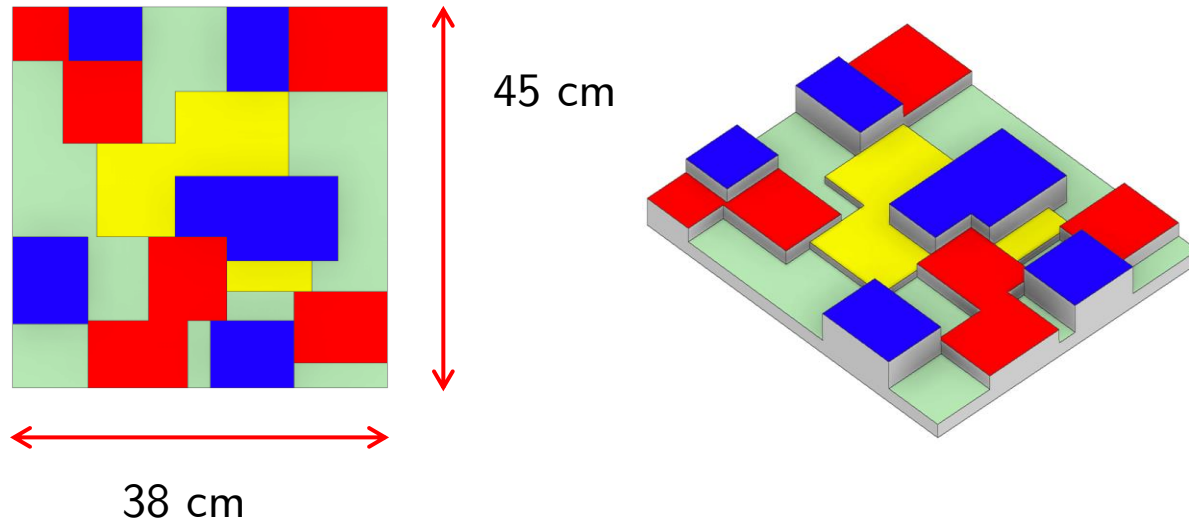
Door Specifications



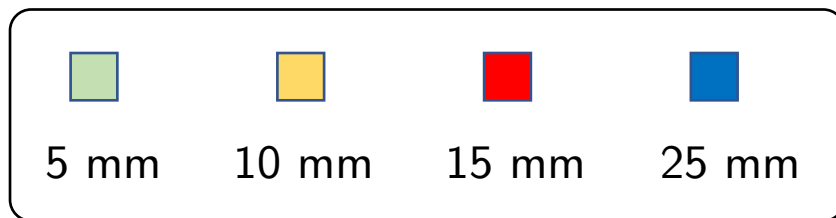
- Door has a handle.
- Teams can measure the handle size and location by themselves.
- Door can be opened in any way possible (**Stage-I**)
- Door needs to be opened using door handle (**Stage-II**)



Uneven Terrain (random steps)



Step Height



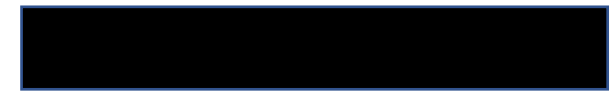
Ditch

Width = 10 cm

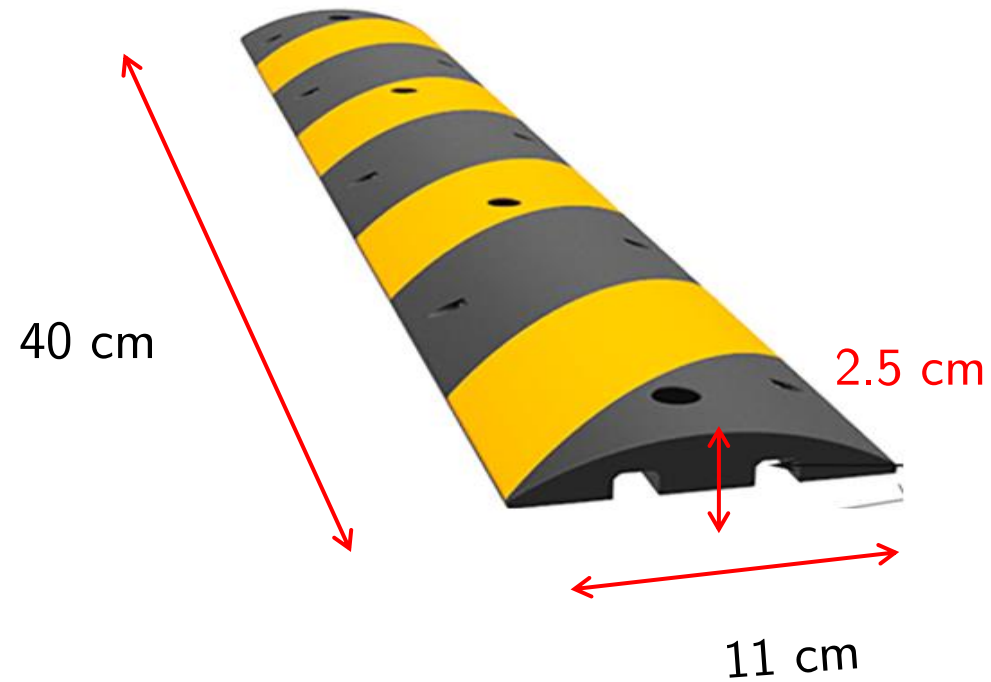


Length = 44 cm

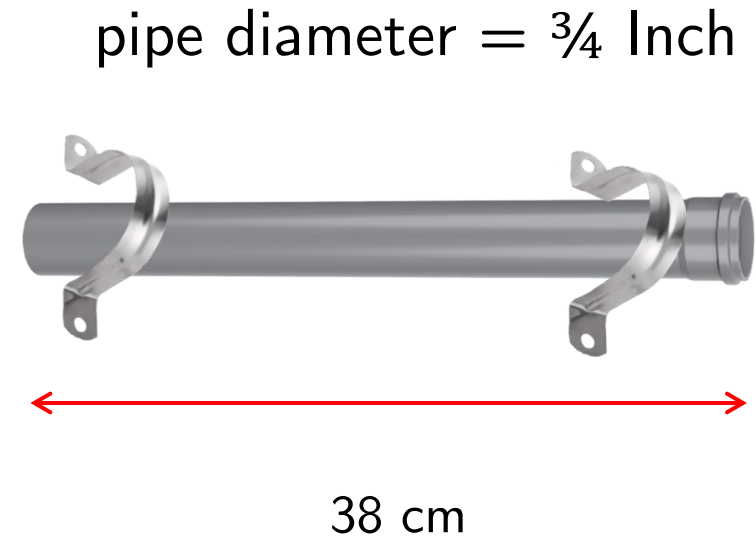
Depth = 4 cm

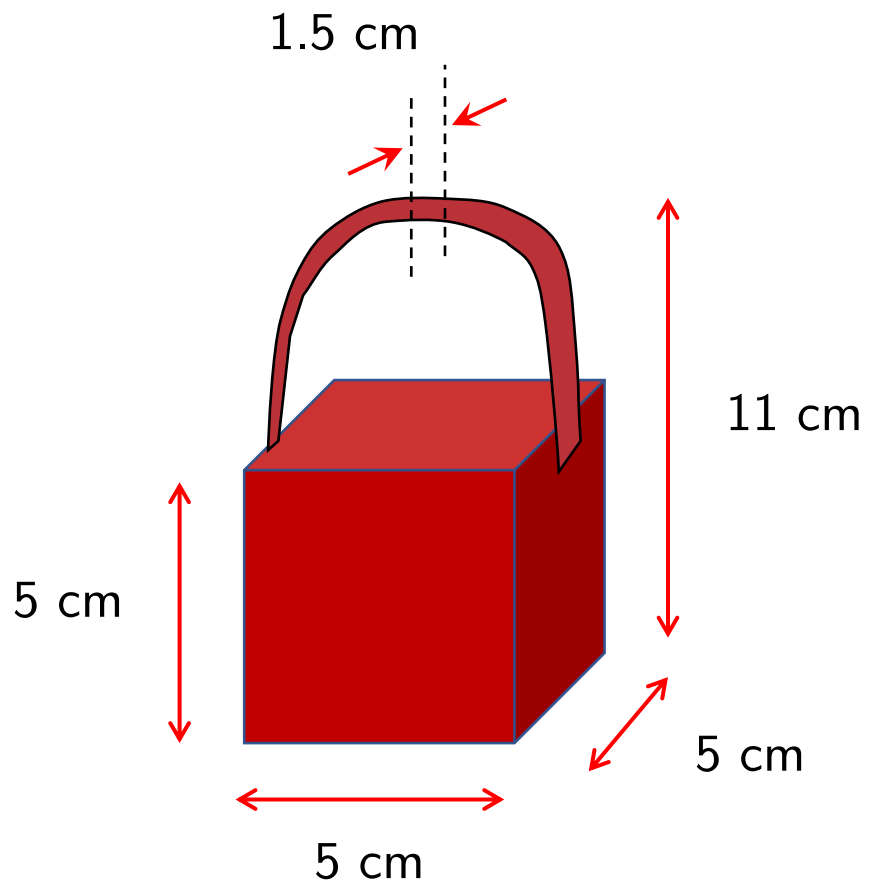


Bump



Roller

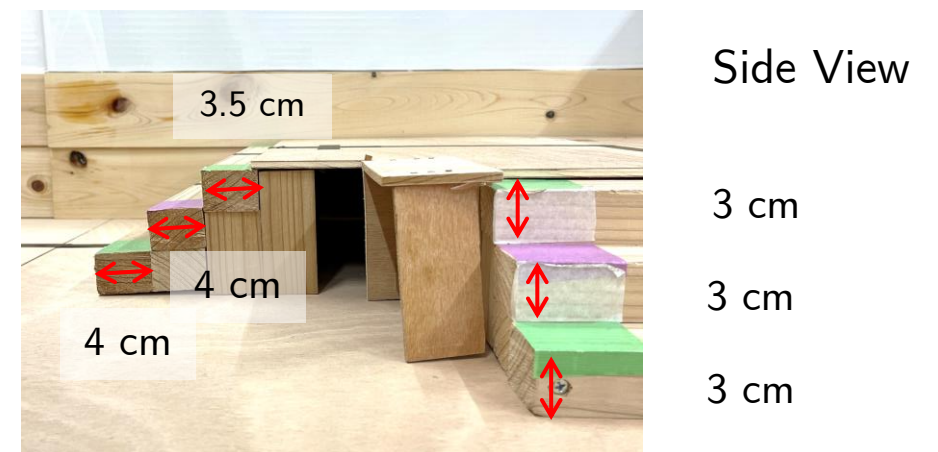
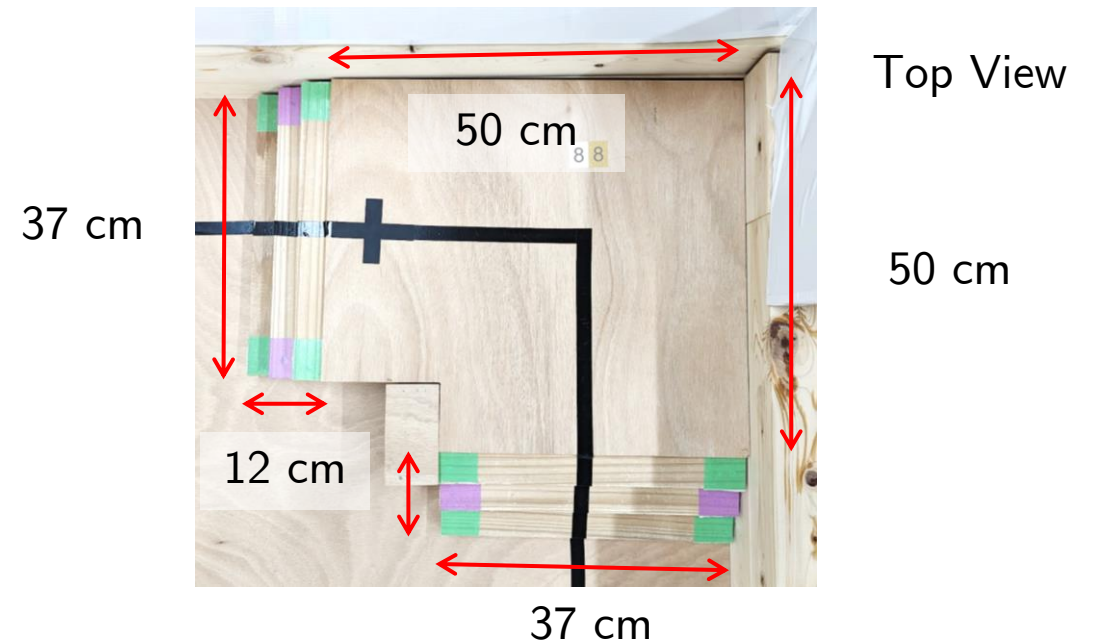




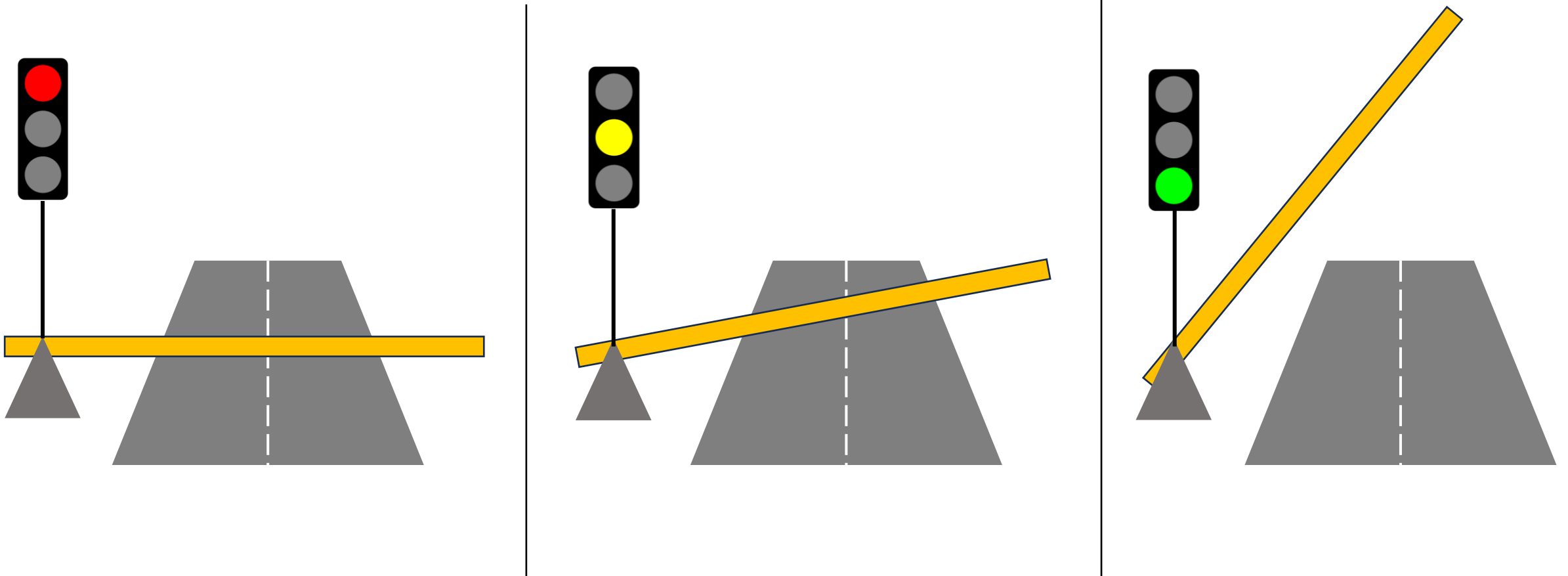
Object

65 cm

Stairs

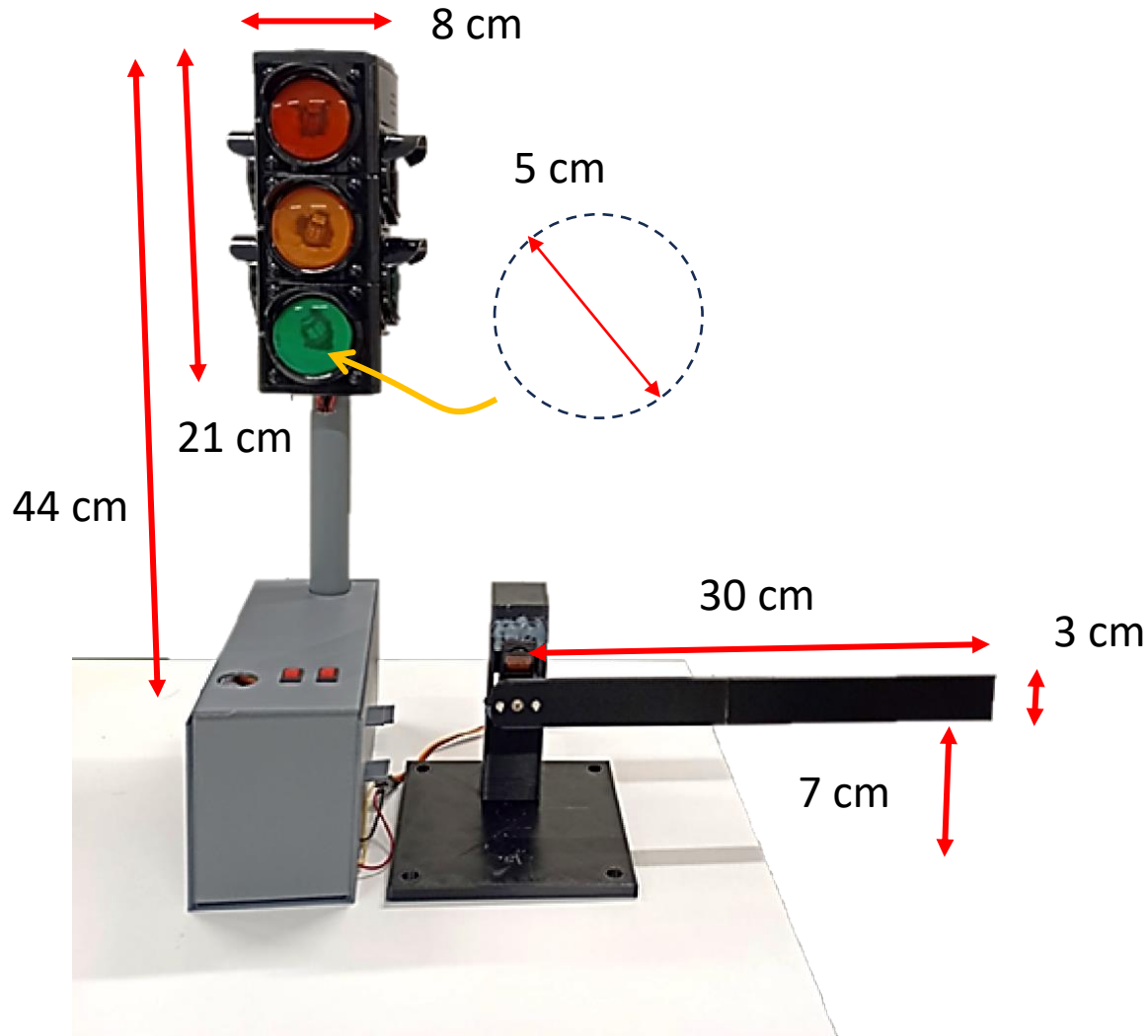


Traffic Light and Barrier



For specifications and operational instructions, see next slide.

Instructions to Open the Barrier



- **Connect to** [redacted] **router** (kuas wlan cannot be used!)
ssid : [redacted]
password: [redacted]
- Set **ROS Domain ID = 1**

Publish the following lowercase strings as ROS2 message when your robot detects traffic light:

- red ← when traffic light is red
- yellow ← when traffic light is yellow
- green ← when traffic light is green

The barrier will open only if the traffic light was **green** and the ROS2 message sent by your robot was '**green**'.

The barrier will close after 2 mins.

Traffic light color changes after every 10 secs.

Traffic Light – Recommended Detection Area

Position A:

Camera height:
15 cm – 30 cm

Position B:

Camera height:
10 cm – 25 cm

Camera should be
placed on the robot

